

MentorMind AI at High Concurrency

Full cost breakdown for the **combined Code2Day + MentorMind platform** on one shared AWS estate, sized for **3,000** and **10,000 truly concurrent users** with headroom for low-latency operation. Per-product attribution in §1.1.

TIER 1 · 3,000 CONCURRENT ₹2.21 lakh_{/mo} \$2,599/month · ~10,000 DAU · ~35,000 registered ₹6.31 per registered student/month	TIER 2 · 10,000 CONCURRENT ₹8.03 lakh_{/mo} \$9,446/month · ~35,000 DAU · ~120,000 registered ₹6.69 per registered student/month
--	--

Two corrections to earlier guidance in this study

- bedrock-mantle is a real, documented AWS endpoint** — AWS's native OpenAI-compatible surface, serving 38 open-weight models including DeepSeek, and **available in ap-south-1 (Mumbai)**. An earlier note in `MENTORMIND_MODULE_COST_BREAKDOWN.md` read "mantle" as a mishearing of Bedrock *Marketplace* and warned of a \$5,176/month per-instance-hour bill. That warning does not apply. Mantle is per-token, serverless, and the correct choice.
- The 23 `response_format` call sites are probably not a migration blocker.** Mantle serves the OpenAI Chat Completions API, so OpenAI parameters are native to it. The earlier estimate of a medium-effort rewrite across eight files should reduce to a base-URL and API-key change. One test call confirms it — see §7.1.

REPORT DATE	9 September 2026 — revision 2
SCOPE	Both products on one shared estate. ~28% of the bill is MentorMind's own, ~4% Code2Day's own, ~68% shared infrastructure. See §1.1
REVISION NOTE	Rev 2 corrects a \$86/month understatement in the Tier 2 summary (S3 audio storage and SQS were omitted from the roll-up row) and adds the \$1.1 per-product attribution. Tier 2 total: \$9,360 → \$9,446 . Tier 1 is unchanged
REGION	ap-south-1 (Mumbai)
FX BASIS	₹85.00 / USD — consistent with <code>AWS_DEPLOYMENT_OPTIONS.md</code>
COMPANION TO	<code>AWS_DEPLOYMENT_OPTIONS.md</code> , <code>PROJECT_DEPLOYMENT_PROFILE.md</code> , <code>MENTORMIND_MODULE_COST_BREAKDOWN.md</code>
PRICE BASIS	Compute, database and cache rates are anchored to us-east-1 published on-demand rates with Savings Plan / Reserved Instance discounts applied as noted. Following the convention of the companion documents, every figure must be re-checked in the AWS Pricing Calculator for ap-south-1 before commitment. AI/API rates are September 2026 list prices with sources in §11.
CONFIDENCE	AI unit rates: VERIFIED AWS unit rates: US-EAST-1 ANCHORED Traffic & token volumes: ASSUMPTION-DRIVEN

1 Executive summary

Scope: this is the whole combined platform, not the MentorMind module alone

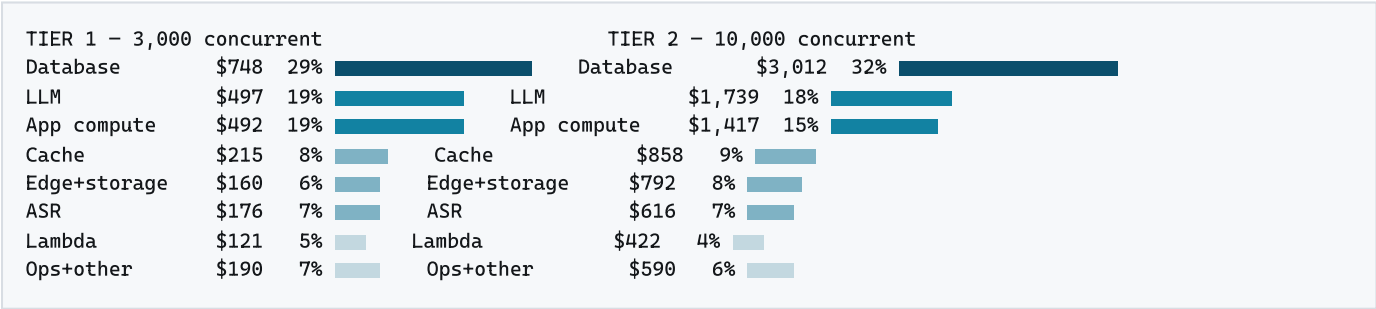
Despite the title, the totals below cost **Code2Day and MentorMind running together** on one shared AWS estate – one application fleet, one database tier, one cache, one CDN. Roughly **two-thirds of the bill is shared infrastructure** that neither product owns. §1.1 breaks the spend into what is attributable to each. If you need a figure for "what does MentorMind add," that is §1.1, not this table.

Both tiers are dominated by three cost centres – the database tier, LLM inference, and application compute. Everything else is rounding.

TABLE 1 — COST SUMMARY BY TIER

Cost centre	Tier 1 (3,000 conc.)	share	Tier 2 (10,000 conc.)	share
Database tier (RDS + Proxy + storage)	\$748	29%	\$3,012	32%
LLM inference (DeepSeek V3.2 via Mantle)	\$497	19%	\$1,739	18%
Application compute (EC2 + ALB)	\$492	19%	\$1,417	15%
Cache tier (ElastiCache)	\$215	8%	\$858	9%
Edge + storage (CloudFront, S3, SQS)	\$160	6%	\$792	8%
Speech-to-text (Groq Whisper turbo)	\$176	7%	\$616	7%
Serverless execution (Lambda)	\$121	5%	\$422	4%
Observability, DNS, backup, NAT	\$150	6%	\$440	5%
AI overflow allowance	\$40	2%	\$150	2%
Total per month	\$2,599	100%	\$9,446	100%
Total per month (INR)	₹2,20,915		₹8,02,910	

Where the money goes



1.1 What is MentorMind's, what is Code2Day's, what is shared

TABLE 1B — COST ATTRIBUTION BY PRODUCT

Attribution	Lines included	Tier 1	Tier 2
MentorMind only	DeepSeek LLM, Groq Whisper, AI overflow, async-speech Lambda, SQS, S3 audio recordings	\$756 29%	\$2,653 28%
Code2Day only	Lambda code execution (the Judge0 replacement)	\$103 4%	\$360 4%
Shared platform	EC2 fleet, ALB, RDS writer + replicas + Proxy + storage, ElastiCache, CloudFront, S3 static, NAT, CloudWatch, Route 53, ECR, Backup	\$1,740 67%	\$6,433 68%
Total		\$2,599	\$9,446

The shared two-thirds cannot be split cleanly — and it is not MentorMind's to claim or avoid

Code2Day at 3,000 or 10,000 concurrent users would need *most* of that shared tier whether MentorMind existed or not: the same ALB, a Multi-AZ database, read replicas, a cache tier, a CDN. So MentorMind's **true marginal cost is not \$756 or \$2,653** — it is those lines plus whatever share of the fleet and database growth its request traffic causes.

MentorMind is the heavier of the two in request terms — dashboards, roadmaps, resume analysis, interview turns and speaking submissions against Code2Day's submit-and-poll coding loop — so a defensible planning split is **roughly 50–60% of the shared tier**. On that basis MentorMind's all-in share is about **\$1,600–1,800/month at Tier 1** and **\$5,900–6,500/month at Tier 2**, with Code2Day carrying the remainder. Splitting it precisely requires the per-endpoint request mix from nginx logs (§10.1), which is unmeasured.

The scaling story is the database, not the AI

From Tier 1 to Tier 2 — a 3.3× jump in concurrency — total cost rises 3.6×, and the database tier rises 4.0×. LLM and ASR scale almost exactly linearly with users and stay a flat 26% of the bill at both tiers. `AWS_DEPLOYMENT_OPTIONS.md` §6 already reached this conclusion at a tenth of the scale: *"The binding constraint at 10,000 is the database, not compute."* At 10,000 concurrent it is the binding constraint by a wide margin, and **two code changes described in §7.2 are prerequisites, not optimisations** — without them Tier 2 cannot be built at any price.

What each tier assumes it must do

"Comfortably handle everything with low latency" is specified concretely in §6 as a latency budget and translated into headroom targets in §3. In summary, both tiers are sized for **50% CPU headroom at peak**, N+1 instance redundancy, Multi-AZ database failover, read replicas serving all dashboard traffic, and an asynchronous speech path that never blocks a web worker.

2 Scale definitions and load model

Every number in this report scales off unmeasured assumptions

`PROJECT_DEPLOYMENT_PROFILE.md`:1530 lists **DAU and peak concurrent users as "UNKNOWN — requires measurement."** The conversions below are industry-standard ratios for timetable-driven campus platforms, not observations of this system. §10.1 gives the two queries that replace them with facts. Until those are run, treat every total as an order-of-magnitude estimate.

TABLE 2 — CONCURRENCY TO POPULATION, AND TO BILLABLE VOLUME

Quantity	Basis	Tier 1	Tier 2
Peak concurrent users	stated requirement	3,000	10,000
Daily active users (DAU)	peak concurrency $\approx$ 30% of DAU	10,000	35,000
Registered students	DAU $\approx$ 29% of registered	35,000	120,000
Peak API request rate	0.2 req/s per concurrent user	600/s	2,000/s
Design target with headroom	$\times 1.5$ for burst and failover	900/s	3,000/s
Graded voice exercises	3 $\times$ $\sim$ 60 s per active user/day	3 min/user/day	3 min/user/day
Voice adoption	40% of DAU record audio on a given day	40%	40%
Monthly audio volume	22 weekday college calendar	264,000 min 4,400 hr	924,000 min 15,400 hr
Monthly LLM calls	after CoE consolidation (§8.2)	$\sim$ 382,000	$\sim$ 1,337,000

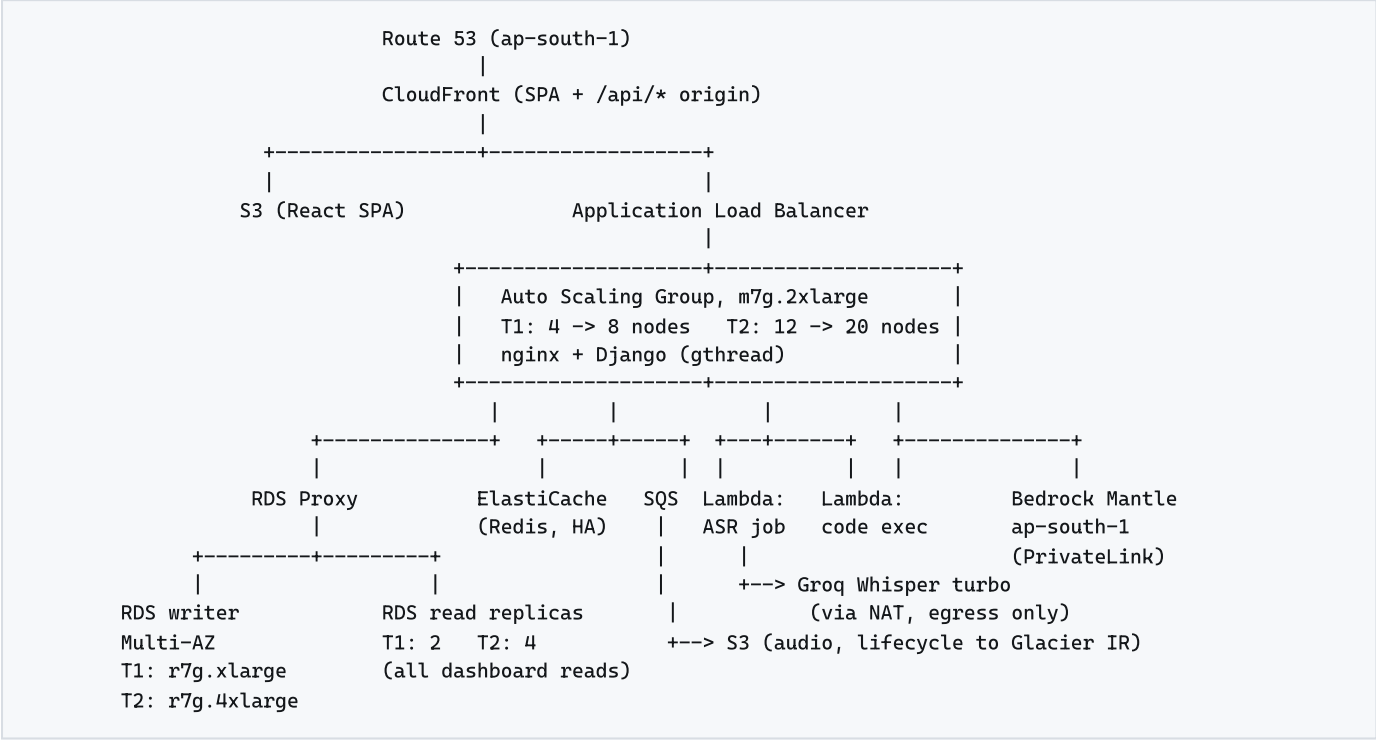
Why 0.2 req/s per concurrent user. A student on a dashboard, a roadmap page, or an interview screen generates one request every $\sim$ 5 seconds counting XHR polling and navigation. This is the single most leveraged assumption in the report: at 0.4 req/s the application compute and database tiers both roughly double. It is directly measurable from nginx logs.

2.1 What is excluded

- **Conversation Partner voice.** Uses the browser's free `SpeechRecognition` API (`ConversationPartner.jsx:160`) and costs nothing server-side. Adopting Bedrock Nova Sonic instead would add **\$16,500/month at Tier 1** and **\$57,750 at Tier 2** at the app's own 15-minute session target — see §9.4.
- **Code2Day's own growth beyond the module.** Judgeo/Lambda code execution is included and scaled, but no new Code2Day features are assumed.
- **Data transfer between AZs** for Multi-AZ replication, which AWS does not charge for RDS Multi-AZ within a Region, and inter-AZ ALB-to-target traffic, budgeted inside the ALB LCU line.
- **Engineering time.** §7 lists the code changes; none are costed in money.

3 Architecture and sizing

Neither tier can run the current single-host shape. Both are sized as an autoscaling application fleet behind an ALB, with the database split into a Multi-AZ writer and dedicated read replicas, and all AI work moved off the request path.



3.1 Application compute sizing

Sizing assumes **40 requests/second per vCPU** for this workload — a blended figure for Django views that are mostly Redis-cached or replica-served, with a minority hitting the writer. This is deliberately conservative against the 25 req/s/vCPU that uncached, join-heavy dashboard views would deliver today, on the assumption that \$7.2's fixes land first.

TABLE 3 — APPLICATION FLEET SIZING

Parameter	Tier 1	Tier 2
Design request rate	900/s	3,000/s
vCPU required at 40 req/s/vCPU	23	75
Instance type	m7g.2xlarge 8 vCPU / 32 GB, Graviton	m7g.2xlarge 8 vCPU / 32 GB, Graviton
Baseline node count (incl. N+1)	4 → 32 vCPU	12 → 96 vCPU
Headroom at design rate	39%	28%
Autoscaling ceiling	8 nodes	20 nodes
Survives loss of	1 node + 1 AZ	2 nodes + 1 AZ

Graviton requires dropping `sentence-transformers`

`requirements.txt` pins `torch==2.10.0+cpu` from the PyTorch CPU index, which publishes **x86_64 wheels only**. Every `m7g` instance above is arm64. The dependency is used in one function — `semantic_matcher.py:24` — which already has a Jaccard fallback. Dropping it is free. Keeping it forces the entire fleet onto x86 (`m7i.2xlarge`), costing roughly **+15% on compute**: about **+\$54/month at Tier 1** and **+\$163 at Tier 2**, plus ~1 GB of RAM per Gunicorn worker.

3.2 Database sizing

The database is the largest line at both tiers and the only component that cannot be scaled by adding cheap nodes, because **the application has no read routing today** (§7.2). Sizing below assumes read routing exists and that all dashboard, leaderboard and analytics traffic is served by replicas.

TABLE 4 — DATABASE TIER SIZING

Component	Tier 1	Tier 2
Writer instance	db.r7g.xlarge 4 vCPU / 32 GB	db.r7g.4xlarge 16 vCPU / 128 GB
Deployment	Multi-AZ	Multi-AZ
Read replicas	2 × db.r7g.large	4 × db.r7g.xlarge
RDS Proxy	Required	Required
Allocated storage (gp3)	200 GB	1,000 GB
Backup retention / PITR	14 days	14 days

RDS Proxy is not optional. `PROJECT_DEPLOYMENT_PROFILE.md:215` records that Code2Day sets no `CONN_MAX_AGE`, so Django opens a **fresh PostgreSQL connection per request**. At Tier 2's 3,000 req/s design rate that is 3,000 connection establishments per second against a database whose practical ceiling is a few thousand *total* connections. Either set `CONN_MAX_AGE` and add RDS Proxy, or the database falls over regardless of instance size. Both, ideally.

3.3 Cache and queue

ElastiCache moves from optional to load-bearing: it holds sessions (today in the PostgreSQL `django_session` table, rewritten on every request per profile §362), the rate limiter (today a process-local dict, `auth_utils.py:195`), bulk-upload progress (today a file-based cache), and the materialised leaderboard from §7.2.

Component	Tier 1	Tier 2
ElastiCache topology	1 primary + 1 replica cache.r7g.large	2 shards × 2 nodes cache.r7g.xlarge
Speech job queue	SQS standard	SQS standard
Speech worker	Lambda, 512 MB	Lambda, 512 MB

4 Tier 1 — 3,000 concurrent users

~10,000 DAU · ~35,000 registered · 900 req/s design target

TABLE 5 — TIER 1 LINE-BY-LINE MONTHLY COST

Line item	Configuration	Pricing basis	USD	INR
APPLICATION COMPUTE				
EC2 application fleet	4 × m7g.2xlarge	3-yr Compute Savings Plan, ~62% off	362	30,770
Application Load Balancer	1 ALB, ~20 LCU avg	\$16.43 fixed + LCU-hours	130	11,050
DATABASE TIER				
RDS writer	db.r7g.xlarge, Multi-AZ	1-yr Reserved, partial upfront	454	38,590
RDS read replicas	2 × db.r7g.large	1-yr Reserved	227	19,295
RDS Proxy	4 vCPU writer	\$0.015 per vCPU-hour	44	3,740
RDS storage + backups	200 GB gp3, 14-day PITR	storage + snapshot overage	23	1,955
CACHE				
ElastiCache for Redis	2 × cache.r7g.large, HA	1-yr Reserved	215	18,275
SERVERLESS EXECUTION				
Lambda — code execution	Code2Day Judge0 replacement	scaled 3.3× from <code>AWS_DEPLOYMENT_OPTIONS.md</code> \$5	103	8,755
Lambda — async speech worker	264k jobs × ~10 s × 512 MB	1.32M GB-s beyond free tier	18	1,530
SQS	~0.8M requests	first 1M free, then \$0.40/M	0	0
EDGE, STORAGE AND NETWORK				
CloudFront	~2 TB egress, India price class	1 TB/mo perpetual free tier, then ~\$0.109/GB	110	9,350
S3 — SPA and media	static assets + resumes	Standard	25	2,125
S3 — audio recordings	~800 GB effective	90-day → Glacier IR, 12-mo delete	25	2,125
NAT Gateway	1 NAT, ~300 GB processed	Groq egress only; Bedrock via PrivateLink	50	4,250
OPERATIONS				
CloudWatch	logs 14-day, metrics, alarms	ingestion + custom metrics	60	5,100
Route 53, ECR, AWS Backup	hosted zone, images, vault	—	40	3,400
Infrastructure subtotal			1,886	1,60,310
AI AND INFERENCE				
DeepSeek V3.2 — Bedrock Mantle	317M in / 162M out tokens	\$0.62/M in, \$1.85/M out	497	42,245
Groq Whisper large-v3-turbo	4,400 audio-hours	\$0.04 per audio-hour	176	14,960
AI overflow allowance	throttle / retry months	~8% contingency	40	3,400
AI subtotal			713	60,605
TIER 1 TOTAL			2,599	2,20,915

Per-student economics. ₹6.31 per registered student per month, or ₹22.09 per daily active user per month. AI is ₹1.73 of the per-student figure.

Attribution. Of the Tier 1 total, **\$756 is MentorMind's own** (DeepSeek, Whisper, the speech queue and audio storage), **\$103 is Code2Day's own** (Lambda code execution), and **\$1,740 is shared platform**. See §1.1.

5 Tier 2 — 10,000 concurrent users

~35,000 DAU · ~120,000 registered · 3,000 req/s design target

TABLE 6 — TIER 2 LINE-BY-LINE MONTHLY COST

Line item	Configuration	Pricing basis	USD	INR
APPLICATION COMPUTE				
EC2 application fleet	12 × m7g.2xlarge	3-yr Compute Savings Plan	1,086	92,310
Application Load Balancer	1 ALB, ~54 LCU avg	bytes-processed dominates LCU	331	28,135
DATABASE TIER				
RDS writer	db.r7g.4xlarge, Multi-AZ	1-yr Reserved, partial upfront	1,815	1,54,275
RDS read replicas	4 × db.r7g.xlarge	1-yr Reserved	907	77,095
RDS Proxy	16 vCPU writer	\$0.015 per vCPU-hour	175	14,875
RDS storage + backups	1 TB gp3, 14-day PITR	storage + snapshot overage	115	9,775
CACHE				
ElastiCache for Redis	2 shards × 2 × cache.r7g.xlarge	cluster mode, 1-yr Reserved	858	72,930
SERVERLESS EXECUTION				
Lambda — code execution	Code2Day Judge0 replacement	scaled 11.7× from \$5 of options doc	360	30,600
Lambda — async speech worker	924k jobs × ~10 s × 512 MB	4.62M GB-s	62	5,270
SQS	~2.8M requests	\$0.40 per million after free tier	1	85
EDGE, STORAGE AND NETWORK				
CloudFront	~6.7 TB egress, India price class	5.7 TB billable at ~\$0.109/GB	621	52,785
S3 — SPA and media	static assets + resumes	Standard	85	7,225
S3 — audio recordings	~2.8 TB effective	90-day → Glacier IR, 12-mo delete	85	7,225
NAT Gateway	2 NAT (multi-AZ), ~1 TB processed	Groq egress only	120	10,200
OPERATIONS				
CloudWatch	logs 14-day, metrics, alarms, dashboards	ingestion scales with fleet	200	17,000
Route 53, ECR, AWS Backup	hosted zone, images, vault	—	120	10,200
Infrastructure subtotal			6,941	5,89,985
AI AND INFERENCE				
DeepSeek V3.2 — Bedrock Mantle	1,110M in / 566M out tokens	\$0.62/M in, \$1.85/M out	1,739	1,47,815
Groq Whisper large-v3-turbo	15,400 audio-hours	\$0.04 per audio-hour	616	52,360
AI overflow allowance	throttle / retry months	~6% contingency	150	12,750
AI subtotal			2,505	2,12,925
TIER 2 TOTAL			9,446	8,02,910

Per-student economics. ₹6.69 per registered student per month, or ₹22.94 per daily active user per month. Notice this is almost identical to Tier 1 — **the architecture scales at roughly constant unit cost**, because the

fixed infrastructure amortises while AI scales linearly. That is the main financial argument for building Tier 1 properly rather than cheaply.

Attribution. Of the Tier 2 total, **\$2,653 is MentorMind's own** (DeepSeek, Whisper, the speech queue and audio storage), **\$360 is Code2Day's own** (Lambda code execution), and **\$6,433 is shared platform**. See §1.1.

6 The latency budget

"Low latency" is only meaningful as a number. The targets below are what the §3 sizing is built to deliver, and each names the component that would break it.

TABLE 7 — LATENCY TARGETS AND WHAT PROTECTS THEM

Interaction	p50	p95	What delivers it
SPA first paint	<0.8 s	<1.8 s	CloudFront edge cache in India; SPA on S3, never the origin
Dashboard API load	<150 ms	<400 ms	Read replica + Redis-cached leaderboard (§7.2 #2)
Login / JWT issue	<120 ms	<300 ms	Redis sessions, RDS Proxy pooling
Resume upload accepted	<300 ms	<800 ms	Direct-to-S3 presigned PUT; parsing deferred to worker
Speech submission accepted	<250 ms	<600 ms	Async path. Request returns a job ID, never waits on Whisper
Speech result available	<6 s	<15 s	Groq turbo (217–228× realtime) + one LLM call
Interview next question	<2.5 s	<6 s	Streaming from Mantle; first token shown, not full response
Roadmap generation	<8 s	<20 s	Async job + progress indicator; long output is unavoidable

6.1 The three things that actually determine latency

1. Synchronous AI calls are the current architecture's fatal flaw

MentorMind's Procfile runs `gunicorn --workers 3` with sync workers; Code2Day runs 12 and, per `PROJECT_DEPLOYMENT_PROFILE.md:1017`, already "collapses at ~20 concurrent submits." A speaking submission today holds one worker for the entire Whisper-plus-LLM round trip — several seconds. At 3,000 concurrent users this does not degrade gracefully; it starves logins and dashboards platform-wide. **Every latency figure in Table 7 assumes the async path in §7.1 #1 exists.** Without it, no amount of instance sizing helps, because the queue forms in Gunicorn rather than in SQS.

2. Region choice, already correct. Both AI providers are reachable with the platform in ap-south-1, but only one is local. `bedrock-mantle.ap-south-1.api.aws` keeps DeepSeek inference in Mumbai — roughly 200 ms of round-trip saved per call versus a US endpoint, on up to 1.34 million calls a month. Groq remains US-hosted, which is tolerable precisely because ASR sits behind the async queue where 200 ms is invisible. This is a real argument for Mantle beyond price.

3. Streaming, not faster models. For interview and roadmap responses, perceived latency is time-to-first-token, not time-to-completion. Mantle supports streaming (`stream: true`). Using it on the two conversational surfaces is worth more than any model downgrade, and costs nothing.

DeepSeek has no published throughput quota on Mantle

The `bedrock-mantle` quota table publishes per-model input/output TPM for **Claude Opus 4.7 only**. For every other model, including DeepSeek, AWS states throughput is *"governed by internal service capacity"* with no per-account quota exposed in Service Quotas. Tier 2 needs roughly **84,000 input TPM sustained and ~250,000 at peak** (\$8.3). That is a material volume to run without a contractual number.

Mantle does have two things in its favour here: **it enforces no requests-per-minute quota at all** — helpful for a workload of 1.34M small calls — and **cached input tokens do not count against the input TPM quota**. Even so, **open an AWS Support case for a published DeepSeek TPM quota before Tier 2 launch**, and note that Provisioned Throughput is *not available* on Mantle, so there is no paid escape hatch if capacity is refused. The fallback is a second model or the `bedrock-runtime` endpoint.

7 Engineering prerequisites

These are not optimisations. The costs in §4 and §5 assume this work is done; several items mean the tier cannot be built at all until it is.

7.1 Module integration

#	Change	Why it gates the estimate	Effort
1	Async speech and resume pipeline — S3 presigned upload → SQS → Lambda → poll or WebSocket	Every Table 7 latency target depends on it. Without it the fleet queues in Unicorn	Large
2	Point the OpenAI SDK at Mantle — <code>OPENAI_BASE_URL=https://bedrock-mantle.ap-south-1.api.aws/v1</code> , Bedrock API key, IAM <code>bedrock-mantle:CreateInference</code>	Probably replaces the feared 23-call-site rewrite with a config change. Verify <code>response_format</code> with one test call first	Small
3	Set <code>store: false</code> on every Mantle request , or set the account data-retention mode to <code>none</code>	See the privacy box below. Not a cost item; a consent item	Trivial
4	Collapse the Committee-of-Experts from 3 LLM calls to 1 (<code>communication_agent.py:125</code> , <code>chat_practice:297</code>)	Worth \$270/mo at T1 and \$945 at T2, and cuts speech latency 3×	Small
5	FileBasedCache → ElastiCache (<code>main/settings.py:110</code>)	Silently returns wrong bulk-upload progress across containers	Small
6	FileSystemStorage → S3 via <code>django-storages</code>	Audio and resumes are destroyed on every redeploy	Small
7	Drop <code>sentence-transformers</code> / <code>torch</code>	Unblocks Graviton; saves \$54/mo at T1, \$163 at T2, plus ~1 GB RAM per worker	Small
8	Fix <code>CORS_ALLOW_ALL_ORIGINS = True</code> (<code>main/settings.py:154</code>) which overrides the allowlist above it, with credentials enabled	Any origin can make credentialed requests. Must precede public exposure	Trivial
9	Make <code>_fallback_transcript()</code> fail loudly (<code>communication_agent.py:377</code>)	Returns a hardcoded sentence that is then scored as real speech. At scale, throttling makes this common	Trivial
10	RAG → <code>pgvector</code> with an index (<code>core/rag/vector_store.py:47</code>)	Currently pulls every chunk into Python per query. Runs on the writer	Medium
11	Rate-limit <code>/api/communication/speaking/</code> per user	An unthrottled endpoint that bills Groq per upload makes the AI bill attacker-controlled	Small
12	Unify Django 5.1.4 → 5.2.1 ; reconcile the two <code>Notification</code> models	Merge hygiene	Small

Privacy: Mantle retains request and response data for 30 days by default

On the Responses API, `store` defaults to `true`, and AWS retains the full input and output for **30 days**. For this platform that means student speech transcripts, resume contents, and interview answers. The data stays in ap-south-1 on the Mantle endpoint and is encrypted at rest, so residency is preserved — but 30-day retention of identifiable student work is a consent question, not a technical one. **Set `store: false` per request, or set the account data-retention mode to `none`, which rejects `store=true` outright.** Decide this before launch, not after an audit.

7.2 Code2Day prerequisites — hard blockers at these tiers

Tier 2 cannot be built until these two land

1. No read routing exists. `AWS_DEPLOYMENT_OPTIONS.md` §6 records that there is "no `DATABASE_ROUTERS` and no `.using()` anywhere, so read replicas are impossible without a code change and you can only scale vertically." The six read replicas across the two tiers in this report — \$227/month and \$907/month of provisioned capacity — **do nothing at all until read routing is implemented.** Without it, the only lever is a bigger writer, and the database line becomes the dominant and eventually unscalable cost.

2. Campus rank is computed in Python on every dashboard load. The options document flags this at 10,000 registered users as "10,000 rows × 3 distinct-count joins, in Python, per dashboard load." At Tier 2's **120,000 registered** it is roughly 12× that, on the writer, at up to 3,000 requests per second. It must be materialised into Redis or a summary table on a schedule.

Also carried forward from the options document and still required: `gthread` workers, per-submission batching for code execution, `--cpus 1` and hard kill-on-timeout on sandboxes, IMDSv2 with hop limit 1, and Phase 0's security fixes — the unauthenticated `DROP DATABASE` endpoint, unauthenticated code execution, and forgeable password-reset token. **None of this report's architecture is safe to expose publicly before Phase 0 is closed.**

8 AI cost derivation

8.1 Speech-to-text

Groq `whisper-large-v3-turbo` at **\$0.04 per audio-hour**, billed per second with a 10-second minimum per request. All recordings here exceed that minimum, so the headline rate holds.

Tier	Minutes/mo	Audio-hours	Rate	Monthly
Tier 1	264,000	4,400	\$0.04/hr	\$176
Tier 2	924,000	15,400	\$0.04/hr	\$616

Two required changes, neither costing anything. Use `speech_fast`, not `speech_best`: `fast_mode` defaults to `False` (`communication_agent.py:13`), nothing passes it, and `views.py:555` hardcodes `speech_best`, so every request runs `large-v3` at ~\$0.111/hr instead of turbo at \$0.04/hr. At Tier 2 that difference is **\$1,093/month** for no benefit that a fluency score can detect. And **request word-level timestamps** via `response_format="verbose_json"`: pacing is currently derived from `file_size / 4000.0` (`communication_agent.py:72`), a byte-size guess, so every words-per-minute penalty is computed off noise.

8.2 Text LLM — DeepSeek V3.2 on Bedrock Mantle

Rate: **\$0.62 per million input tokens, \$1.85 per million output tokens**. Context window 163,840 tokens. Per-token and serverless — no instance-hour commitment, and no Provisioned Throughput available or required.

Volumes below assume the Committee-of-Experts consolidation (§7.1 #4), a keyword router for the load balancer, and trimmed JSON responses. The un-optimised figures are 1.9× higher.

TABLE 8 — TOKEN VOLUME PER MONTH, TIER 1 (MULTIPLY BY 3.5 FOR TIER 2)

Agent	Calls	in/call	out/call	Input (M)	Output (M)
Speaking evaluation (consolidated)	264,000	400	250	105.6	66.0
Interview Q&A and evaluation	88,000	1,500	600	132.0	52.8
Resume analysis	22,000	3,000	1,200	66.0	26.4
Roadmap generation	6,700	2,000	2,500	13.4	16.8
Load-balancer routing	0	moved to the existing keyword fallback, <code>load_balancer_agent.py:47</code>			
Total	380,700			317.0	162.0

Tier	Input	Output	Input cost	Output cost	Monthly
Tier 1	317 M	162 M	\$196	\$300	\$497
Tier 2	1,110 M	566 M	\$688	\$1,047	\$1,739

Output is 60% of the LLM bill at 3× the input rate, because every agent returns structured JSON scores plus prose feedback. Shortening responses is worth three times as much as shortening prompts — the opposite of most cost-optimisation intuition.

8.3 Throughput check

Tier 2, spread over 22 weekdays x 10 peak hours = 13,200 peak-minutes/month

input

1,110 M / 13,200 = 84,000 input TPM sustained

output

566 M / 13,200 = 43,000 output TPM sustained

peak factor x3

= 252,000 input TPM / 129,000 output TPM at peak

requests

1.34 M / 13,200 = 101 requests/minute sustained (no RPM quota on Mantle)

This is the volume referenced in the §6.1 quota warning. Prompt caching helps twice over — it reduces the input bill, and cached reads do not count against the input TPM quota at all.

8.4 Alternatives priced at these tiers

Option	Tier 1	Tier 2	Note
Groq <code>gpt-oss-20b/120b</code>	\$150	\$525	Cheapest, but US-hosted with unpublished limits and no IAM
DeepSeek V3.2 — Mantle CHOSEN	\$497	\$1,739	ap-south-1, IAM, OpenAI-compatible, 164k context
Claude Haiku 4.5 — Bedrock	\$1,127	\$3,940	2.3× DeepSeek; hold as the fallback if JSON reliability disappoints
Amazon Transcribe instead of Groq	\$6,336	\$22,176	36× Groq for less capability than Deepgram
Deepgram Nova-3 instead of Groq	\$1,135	\$3,973	Buys native filler-word detection — see §9.4

9 Cost levers and options

9.1 Applied in the totals above

Lever	T1 saving	T2 saving
3-year Compute Savings Plan on the application fleet	\$829	\$2,487
1-year Reserved Instances on RDS and ElastiCache	\$471	\$1,881
Committee-of-Experts collapsed 3 calls → 1	\$270	\$945
Whisper turbo instead of <code>large-v3</code>	\$312	\$1,093
Trimmed JSON output payloads	\$150	\$525
Keyword router replacing LLM intent classification	\$53	\$186
Graviton across fleet, cache and database	\$54	\$163
Audio lifecycle to Glacier IR, 12-month delete	\$50	\$175
PrivateLink for Bedrock, avoiding NAT data charges	\$18	\$62
Total already applied	\$2,207	\$7,517

Without these, Tier 1 would cost about **\$4,806/month (₹4.09 lakh)** and Tier 2 about **\$16,963/month (₹14.42 lakh)**. The optimisations are worth more than the remaining bill.

9.2 Available but not taken

Lever	T1	T2	Why not counted
Bedrock prompt caching	−\$100	−\$350	Mantle supports it and cached reads skip the TPM quota, but these system prompts are short (~200–500 tokens) and may fall under the minimum cacheable prefix. Test it; treat as upside
3-year RDS Reserved Instances	−\$150	−\$600	A 3-year commitment on the component most likely to be resized. 1-year is the prudent term here
Aurora PostgreSQL with reader endpoints	varies	varies	Better read scaling and failover than RDS replicas at Tier 2, but pricing depends on I/O volume, which is unmeasured. Evaluate with real numbers
CloudFront price class restriction	−\$20	−\$120	Excluding distant edge locations is fine for an India-only audience. Take it if the audience is confirmed domestic
Fargate Spot for async workers	−\$8	−\$28	Lambda is already cheaper at this duty cycle
Bedrock Batch (50% off)	—	—	Not available on Mantle. Batch inference applies only to <code>bedrock-runtime</code>

9.3 AWS Educate / Academy credits

`AWS_DEPLOYMENT_OPTIONS.md` §8 #2 recommends applying for institutional credits, noting they *"could plausibly cover a year of either build outright."* At these tiers that is **₹26.5 lakh/year at Tier 1** or **₹96.3 lakh/year at Tier 2** — almost certainly beyond an educational credit grant, but large enough that even partial coverage is worth the application. Treat credits as a discount on a real budget, never as the budget. And per §8 #1 of that document, **open a Paid plan account**; the Free plan auto-closes after six months.

9.4 Deliberately excluded, with prices

Option	T1/mo	T2/mo	Verdict
Bedrock Nova Sonic for Conversation Partner	\$16,500	\$57,750	NO At the app's own 15-minute session target this exceeds the entire rest of the platform by 6×. It replaces something that is currently free. If ever adopted: pilot it, measure real token consumption, and hard-cap session length first
Deepgram Nova-3 for filler words	+\$959	+\$3,357	DEFER The one genuine capability gap. Whisper strips "um"/"uh" at every size, so the filler counter at <code>communication_agent.py:76</code> only ever matches "like/so/actually/basically"
RDS Multi-AZ read replicas	+\$227	+\$907	DEFER Replicas are already in separate AZs; making each one Multi-AZ is belt-and-braces
Self-hosted Whisper on GPU	\$450+	\$1,400+	NO More expensive than Groq at both tiers, plus GPU operations

10 Risks, sensitivities and unverified inputs

10.1 Measure these first

Two queries convert this report from an estimate into a plan. They are the highest-value hour of work available anywhere in this study.

```
-- Daily active users
SELECT count(*) FROM student_profiles WHERE last_login_on = CURRENT_DATE;

-- Peak concurrency, from the nginx access log
awk '{print $4}' code2day.access.log | uniq -c | sort -rn | head

-- And instrument generate_completion() to log real token counts for one week.
```

10.2 Sensitivity

If this assumption is wrong	Effect on Tier 2 total
Requests per concurrent user is 0.4/s, not 0.2/s	+\$2,900/mo — fleet and database both roughly double
Voice adoption is 100% of DAU, not 40%	+\$3,540/mo
Token counts per call are 2× the estimate	+\$1,739/mo
DeepSeek reasoning mode is left enabled	+\$2,000 to \$9,400/mo — see the box below
ap-south-1 carries a 20% uplift over us-east-1 rates	+\$1,370/mo
Read routing is never implemented	Replicas become dead spend; vertical writer scaling only. Tier 2 is not reachable

The largest single cost risk: reasoning mode

DeepSeek V3.2 is a reasoning-capable mixture-of-experts model, and reasoning tokens bill as **output at \$1.85/M**. Every task in this platform is structured scoring at `temperature=0.0` and needs no chain-of-thought. Left enabled across 1.34 million calls a month, output tokens can rise 3–10×, taking the Tier 2 output line from **\$1,047 to between \$3,100 and \$10,470** — silently, with no error and no visible change in the product. **Pin it off explicitly and assert it in a test.**

10.3 Unverified inputs

Item	Status and what to do	
DAU and peak concurrency	UNKNOWN	PROJECT_DEPLOYMENT_PROFILE.md:1530 . Run \$10.1
ap-south-1 unit prices	ANCHORED TO US-EAST-1	The companion document already warns that aggregators disagree materially on Mumbai. Use the AWS Pricing Calculator
DeepSeek V3.2 price in ap-south-1	US RATE	\$0.62/\$1.85 is the published US East/West rate. Confirm in the Bedrock console
response_format on Mantle	INFERRED	Mantle serves the OpenAI Chat Completions API, so it should be native. One test call decides whether \$7.1 #2 is trivial or a rewrite
DeepSeek reasoning-mode default on Mantle	UNCONFIRMED	Determines a swing of up to \$9,400/mo at Tier 2. Verify, then pin
DeepSeek TPM capacity on Mantle	NO PUBLISHED QUOTA	Open an AWS Support case before Tier 2. No Provisioned Throughput escape hatch exists on this endpoint
Requests per second per vCPU	ESTIMATED AT 40	Load-test with k6 or Locust from inside the Region, as profile \$2271 recommends
Audio bitrate	ASSUMED ~1 MB/MIN	Measure actual SpeakingSession.audio_file sizes; S3 and NAT scale off it
CloudFront egress volume	ESTIMATED	2 TB and 6.7 TB per month. Depends heavily on SPA bundle size and cache hit ratio
torch on aarch64	STATED AS FAILING	The PyTorch CPU index publishes x86_64 wheels only. Confirm with a pip install on a t4g instance

11 Sources

AWS documentation. Responses API — bedrock-mantle endpoint (regions, authentication, store retention behaviour) · Quotas for the bedrock-mantle endpoint (input/output TPM, no RPM enforcement, cached tokens exempt, no Provisioned Throughput) · DeepSeek V3.2 model card · DeepSeek, OpenAI and Qwen models in additional Regions (ap-south-1) · Amazon Bedrock pricing · AWS Lambda pricing · Amazon RDS for PostgreSQL pricing · EC2 on-demand pricing · Amazon SageMaker AI pricing.

Rate references. DeepSeek V3.2 on Bedrock — \$0.62/\$1.85 per 1M, 163,840 context · Groq pricing 2026 — Whisper turbo \$0.04/audio-hour · Groq gpt-oss-20b — \$0.075/\$0.30 per 1M · Claude Haiku 4.5 — \$1/\$5 per 1M · Amazon Transcribe — \$0.024/min Tier 1 · Deepgram Nova-3 — \$0.0043/min batch · Nova 2 Sonic — \$3/\$12 per 1M speech tokens · RDS db.r7g instance rates · Bedrock billing shapes and Mantle service tiers.

Repository documents. AWS_DEPLOYMENT_OPTIONS.md §§1, 5, 6, 8, 9 · PROJECT_DEPLOYMENT_PROFILE.md §§215, 286, 362, 1017, 1530, 2179, 2271 · MENTORMIND_MODULE_COST_BREAKDOWN.md §§2, 4, 5, 7, 9, 10.

MentorMind AI at High Concurrency — cost study for the Learn2Lead module on Code2Day. Prepared 9 September 2026 for ap-south-1 at ₹85/USD. Compute, database and cache rates are us-east-1 anchored; AI rates are September 2026 list prices. **Re-verify all figures in the AWS Pricing Calculator before commitment.** Companion to AWS_DEPLOYMENT_OPTIONS.md , PROJECT_DEPLOYMENT_PROFILE.md and MENTORMIND_MODULE_COST_BREAKDOWN.md .